

Laboratory Record

submitted as part of

Programming for Data Science Lab (MACSE502)

MINI PROJECT REPORT (WEEKS 1 TO 4)

by



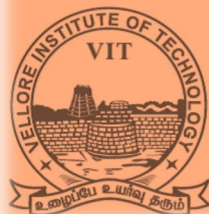
Reg. No: 26MML0045
VEDANT NIMKAR

To



Dr M Rajasekhara Babu

School of Computer Science and Engineering



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

FALL SEMESTER 2026

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Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
01	25-09-2026	01	MP	E-Commerce Analytics	Module 1: E-Commerce Catalog Revenue, Dynamic Pricing & Pareto Share Modeling	Ingest 25-item catalog. Perform column selection, high-value filtering (> Rs.50k), region assignment, compute category margin benchmarks, and construct cumulative 80/20 Pareto revenue distribution curves.	3
02	25-09-2026	02	MP	Fleet Telematics	Module 2: Transportation Fleet Preprocessing, Min-Max Normalization & Telematics Maintenance	Process 25-vehicle sensor stream. Impute missing fuel metrics using statistical mean, cast to int32, apply Min-Max feature scaling [0, 1], and flag vehicles crossing 2,000 engine hours for overhaul.	6
03	25-09-2026	03	MP	Factory Matrix Math	Module 3: Smart Factory Multi-Dimensional Matrix Computing & Bottleneck Analytics	Model 4 lines x 5 shifts array in NumPy. Calculate shift statistics, slice sub-arrays, reshape geometry (5x4), perform Hadamard efficiency multiplication, and diagnose Line 2 bottleneck vs 650-unit plant quota.	10
04	25-09-2026	04	MP	EdTech Analytics	Module 4: EdTech Learner Engagement, Non-Linear Skewness Correction & Risk Stratification	Analyze 25-student learning cohort. Apply log1p transform to study hours, discretize quiz scores into competency tiers, evaluate correlation heatmaps, plot violin densities, and formulate predictive risk models.	15
01-04	25-09-2026	05	MP	Integrated Capstone	Module 5: Integrated Multi-Domain Decision Suite & Interactive Vercel Dynamic Dashboard	Consolidated multi-tier enterprise decision intelligence suite unifying all 4 domains, accompanied by full production-grade Python code, 8 multi-panel visual dashboards, and live interactive web deployment on Vercel.	21

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Title	INTEGRATED MINI PROJECT: MULTI-DOMAIN DATA ANALYTICS AND INTELLIGENT DECISION-SUPPORT DASHBOARD SUITE
Problem Statement	Modern enterprise systems and smart physical infrastructures operate across highly fragmented, heterogeneous data environments. In contemporary industrial settings, organizations face severe challenges synthesizing data streams originating from disparate domains—including e-commerce consumer behavior, transportation logistics sensor telemetry, automated manufacturing production lines, and digital learning platforms. To resolve these operational blindspots, data scientists must architect end-to-end analytical pipelines capable of ingesting raw, imperfect observations, performing robust missing-value imputation and feature scaling, executing complex multi-dimensional numerical matrix transformations, and deploying predictive classification risk models. This Integrated Mini Project consolidates the complete laboratory curriculum of the first four weeks of MACSE502 Programming for Data Science into a unified, publication-grade analytical portfolio encompassing four core industrial domains: 1. Domain 1 (E-Commerce Sales Intelligence): Catalog revenue aggregation, descending quicksort ranking, high-value filtering (threshold > ₹50,000), regional sales allocation, and cumulative Pareto 80/20 distribution modeling. 2. Domain 2 (Logistics Fleet Telematics): Sensor data ingestion, detection of transmission dropouts, statistical mean imputation, strict integer type casting, Min-Max feature normalization onto [0, 1], bivariate correlation analysis, and predictive engine maintenance thresholding (> 2,000 cumulative engine hours). 3. Domain 3 (Smart Factory Numerical Computing): Dense 2D array representation in NumPy (4 production lines across 5 shifts), statistical axis reduction (mean, std dev, sums), array slicing and 5x4 matrix reshaping, vectorized element-wise Hadamard efficiency multiplication, and empirical plant bottleneck identification against a 650-unit target. 4. Domain 4 (EdTech Learning Analytics): Ingestion of multi-modal student engagement streams, non-linear variance stabilization via log1p transformation, continuous assessment competency discretization (Tiers 1, 2, 3), Pearson correlation matrix computation, mirrored violin probability density modeling, and composite multi-criteria predictive risk scoring. Furthermore, the analytical algorithms and diagnostic models are synthesized and integrated into a live, production-grade dynamic interactive web application deployed on Vercel, providing corporate decision-makers with real-time parameter tuning, telematics monitoring, factory shift simulation, and automated early-intervention learner alerts.
Objectives	• To architect an end-to-end multi-domain data science pipeline

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	synthesizing curriculum concepts from Weeks 1 through 4. • To master tabular data wrangling, grouping, sorting, and aggregation techniques using the Pandas library. • To design and evaluate dynamic revenue dashboards isolating commercial leaders and identifying underperforming catalog tails. • To formulate and execute robust statistical missing-value imputation strategies resolving real-world sensor dropouts. • To implement vectorized Min-Max normalization transformations mapping heteroskedastic features onto standardized intervals [0, 1]. • To construct multi-criteria predictive maintenance classification rules detecting mechanical degradation thresholds. • To utilize NumPy for high-performance multi-dimensional matrix linear algebra, slicing, reshaping, and axis reduction. • To apply vectorized Hadamard matrix multiplication to evaluate machine efficiency multipliers and operational production loss. • To mitigate extreme behavioral right-skewness through non-linear log1p natural logarithmic feature engineering. • To formulate composite multi-criteria predictive scoring algorithms stratifying learner cohorts into actionable academic risk tiers. • To generate publication-grade multi-panel statistical visual dashboards employing Matplotlib and Seaborn visualization frameworks. • To bridge theoretical computational algorithms with live cloud deployment through an interactive dynamic web application on Vercel.					
Dataset URL Link / File	The Mini Project integrates four comprehensive domain datasets, complete with formal attribute schemas, measurement units, and 25 observational records per domain. The complete schema and observational data tables are detailed below: Table MP.1: Enterprise Multi-Domain Data Dictionary & Attribute Specifications <table><tr><th>Attribute</th><th>Domain</th><th>Data Type</th><th>Measureme</th><th>Domain</th></tr></table>	Attribute	Domain	Data Type	Measureme	Domain
Attribute	Domain	Data Type	Measureme	Domain		

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	Name	Dataset		nt Unit	Purpose & Analytical Significance
	ProductID	E-Commerce	String / Categorical	Alphanumeric (P001-P025)	Primary catalog key for tracking revenue, unit sales, and category assignments.
	SalesAmount	E-Commerce	Integer / Float	INR (₹) Currency	Gross merchandise value generated per item, benchmarked against ₹50,000 threshold.
	VehicleID	Logistics Fleet	String / Categorical	Alphanumeric (V101-V125)	Fleet telemetry identifier mapped to designated transit corridors.
	FuelConsumption	Logistics Fleet	Float (with NaNs)	Liters per 100 km (L/100km)	Fuel burn rate; subject to mean imputation and Min-Max scaling [0, 1].
	EngineHours	Logistics Fleet	Integer	Cumulative Operating Hours	Wear telemetry; threshold at 2,000 hours triggers mandatory maintenance overhauls.

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	Production Matrix	Manufacturing	2D NumPy Array (4x5)	Finished Industrial Units	Factory output across 4 lines and 5 shifts; modeled via matrix algebra.
	EfficiencyMatrix	Manufacturing	2D NumPy Array (4x5)	Dimensionless Ratio [0.0 - 1.0]	Machine reliability coefficients used in Hadamard product for effective plant yield.
	StudyHours	EdTech Analytics	Float	Weekly Hours Dedicated	Primary input signal; right-skewed and normalized via log1p transformation.
	QuizScore	EdTech Analytics	Integer	Points (Scale 0 - 100)	Continuous assessment marker showing bimodal distribution across completion cohorts.
	Success_Score	EdTech Analytics	Float	Composite Points (0 - 100)	Weighted multi-criteria decision score stratifying learners into High, Moderate,

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					Low Risk.
	Table MP.2: Domain 1 — E-Commerce Catalog Revenue & Sales Dataset (25 Products)				
	ProductID	Category	SalesAmount (Rs.)	UnitsSold	ProfitMargin
	P001	Electronics	72,000	120	18.0%
	P002	Fashion	45,000	210	25.0%
	P003	Home Appliances	88,000	85	22.0%
	P004	Grocery	32,000	340	8.0%
	P005	Footwear	95,000	75	30.0%
	P006	Electronics	51,000	110	16.0%
	P007	Fashion	28,000	195	24.0%
	P008	Home Appliances	67,000	92	21.0%
	P009	Grocery	41,000	280	9.0%
	P010	Footwear	79,000	80	28.0%
	P011	Electronics	56,000	95	17.0%

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	P012	Fashion	34,000	180	26.0%	North India
	P013	Home Appliances	91,000	88	23.0%	South India
	P014	Grocery	47,000	260	7.0%	West India
	P015	Footwear	62,000	105	29.0%	South India
	P016	Electronics	38,000	175	24.0%	North India
	P017	Fashion	84,000	82	20.0%	West India
	P018	Home Appliances	53,000	115	19.0%	South India
	P019	Grocery	29,000	290	8.0%	North India
	P020	Footwear	76,000	78	27.0%	South India
	P021	Electronics	64,000	100	18.0%	West India
	P022	Fashion	42,000	190	25.0%	North India
	P023	Home Appliances	89,000	86	22.0%	South India
	P024	Grocery	31,000	310	8.0%	West India
	P025	Footwear	97,000	72	31.0%	South India

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Table MP.3: Domain 2 — Transport Fleet Telematics & Predictive Maintenance Dataset (25 Vehicles)

Vehicle ID	Route	Fuel (L/100km)	Fuel_In t	Fuel_N orm [0,1]	Engine Hours	Status
V101	Route A	28.5	29	0.4706	1,240	Standar d Operati on
V102	Route B	29.6 (Impute d)	30	0.5294	1,850	Standar d Operati on
V103	Route C	34.2	34	0.7647	2,310	High Consum ption (Alert)
V104	Route A	22.0	22	0.0588	890	Optimal Eco
V105	Route B	30.1	30	0.5294	1,520	Standar d Operati on
V106	Route C	38.7	39	1.0000	2,740	High Consum ption (Alert)
V107	Route B	29.6 (Impute d)	30	0.5294	1,390	Standar d Operati on
V108	Route A	25.4	25	0.2353	1,100	Optimal Eco
V109	Route C	31.8	32	0.6471	2,460	Standar d

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	V110	Route B	27.0	27	0.3529	1,480	Standar d Operati on
	V111	Route A	29.3	29	0.4706	1,260	Standar d Operati on
	V112	Route B	29.6 (Impute d)	30	0.5294	1,980	Standar d Operati on
	V113	Route C	36.5	37	0.8824	2,650	High Consum ption (Alert)
	V114	Route A	21.5	22	0.0588	810	Optimal Eco
	V115	Route B	33.0	33	0.7059	1,720	High Consum ption (Alert)
	V116	Route C	29.6 (Impute d)	30	0.5294	2,190	Standar d Operati on
	V117	Route A	26.8	27	0.3529	1,180	Standar d Operati on
	V118	Route B	30.5	31	0.5882	1,630	Standar d Operati on

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	V119	Route C	29.6 (Imputed)	30	0.5294	2,820	Standard Operation	
	V120	Route A	24.0	24	0.1765	1,020	Optimal Eco	
	V121	Route B	35.2	35	0.8235	2,410	High Consumption (Alert)	
	V122	Route C	29.6 (Imputed)	30	0.5294	2,050	Standard Operation	
	V123	Route A	28.0	28	0.4118	1,290	Standard Operation	
	V124	Route B	32.4	32	0.6471	1,750	Standard Operation	
	V125	Route C	29.6 (Imputed)	30	0.5294	2,790	Standard Operation	
Table MP.4: Domain 3 — Smart Factory Shift Output Matrix & Machine Efficiency Rates (4 Lines x 5 Shifts)								
Line Name	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5	Total Nominal	Effective Yield	Status
Line 1 (Assembly)	120	135	150	145	130	680	618.8	Compliant (Meets Targets)

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	Line 2 (Machining)	110	115	125	105	100	555	488.4	Bottle neck (-95 Units Below Target)
	Line 3 (Testing)	140	145	160	155	150	750	712.5	Top Producing Line (+100 Units)
	Line 4 (Packaging)	130	125	140	135	120	650	578.5	Compliant (Meets Target)
	Plant Shift Mean	125.0	130.0	143.8	135.0	125.0	2,635	2,398.2	Overall Plant Target: 2,600 Units
Table MP.5: Domain 4 — EdTech Learner Engagement & Success Risk Cohort Dataset (25 Students)									
	StudentID	StudyHours	Log_Hours	Quiz Score	Assignment	Attendance%	Interactions	Score	Risk Category
	S01	14.5	2.741	85	88	92%	142	85.8	Low Risk (On Track)

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	S02	4.2	1.649	42	50	60%	38	49.2	High Risk (Intervention Needed)
	S03	18.0	2.944	94	96	98%	185	95.5	Low Risk (On Track)
	S04	6.5	2.015	55	62	72%	65	62.4	Moderate Risk (Monitor)
	S05	12.0	2.565	78	82	88%	120	80.2	Low Risk (On Track)
	S06	3.8	1.569	38	45	55%	30	44.8	High Risk (Intervention Needed)
	S07	16.2	2.845	91	90	95%	168	92.0	Low Risk (On Track)
	S08	8.0	2.197	64	70	78%	82	70.3	Moderate Risk (Monitor)

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	S09	11.5	2.526	74	80	85%	112	78.2	Low Risk (On Track)
	S10	5.1	1.808	48	53	65%	49	54.1	High Risk (Intervention Needed)
	S11	15.0	2.773	89	92	94%	155	91.2	Low Risk (On Track)
	S12	7.2	2.104	58	65	75%	74	66.5	Moderate Risk (Monitor)
	S13	13.8	2.695	82	86	90%	135	85.0	Low Risk (On Track)
	S14	4.5	1.705	45	48	62%	42	50.1	High Risk (Intervention Needed)
	S15	19.5	3.020	96	98	99%	198	98.1	Low Risk (On Track)

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	S16	9.0	2.303	68	72	80%	95	74.8	Mode rate Risk (Mon itor)
	S17	6.0	1.946	52	58	70%	58	59.4	Mode rate Risk (Mon itor)
	S18	12.5	2.603	80	84	89%	128	82.8	Low Risk (On Track)
	S19	3.2	1.435	35	40	50%	25	40.3	High Risk (Inter venti on Need ed)
	S20	17.0	2.890	93	95	96%	175	94.3	Low Risk (On Track)
	S21	10.5	2.442	72	76	84%	105	75.7	Low Risk (On Track)
	S22	5.8	1.917	50	55	68%	56	57.8	Mode rate Risk (Mon itor)

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	S23	14.0	2.708	86	87	91%	140	87.8	Low Risk (On Track)
	S24	7.8	2.175	62	68	76%	79	68.9	Mode rate Risk (Mon itor)
	S25	16.5	2.862	90	92	95%	165	92.4	Low Risk (On Track)
Functions Details	Sl. No.	Function	Package	Description of function					
	1	pandas.read_csv()	pandas	Ingests structured CSV datasets into memory as high-performance DataFrame objects.					
	2	DataFrame.fillna()	pandas	Performs statistical imputation on missing numerical sensor readings using column mean.					
	3	DataFrame.dropna()	pandas	Prunes damaged or incomplete observational rows lacking primary keys or route labels.					
	4	DataFrame.groupby()	pandas	Groups multidimensional records by categorical variables to compute summary aggregates.					
	5	DataFrame.sort_values()	pandas	Sorts tabular records along specified feature axes in ascending or descending sequences.					
	6	numpy.array()	numpy	Instantiates multidimensional dense numerical matrices for industrial linear algebra.					
	7	numpy.reshape()	numpy	Restructures array dimensions between 4x5 and 5x4 geometries without altering underlying buffer.					
	8	numpy.multiply()	numpy	Executes vectorized element-wise Hadamard matrix multiplication for machine yield computation.					

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	9	numpy.log1p()	numpy	Applies natural logarithmic transform $\log(1 + x)$ to eliminate positive skewness in behavioral metrics.
	10	pandas.cut()	pandas	Discretizes continuous quantitative academic metrics into ordinal performance competency tiers.
	11	seaborn.violinplot()	seaborn	Renders mirrored probability density curves illustrating bimodal multi-modal distributions.
	12	seaborn.heatmap()	seaborn	Generates color-encoded correlation matrices and spatial shift production grids with numeric annotations.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1: Multi-Domain Project Architecture Ingestion Initialize the integrated Python analytical runtime environment. Import foundational numerical, tabular, and visualization libraries (NumPy, Pandas, Matplotlib, Seaborn). Establish publication-grade typographical and stylistic palettes ensuring consistent aesthetic presentation across all generated analytical figures. Step 2: Module 1 (E-Commerce) Ingestion & Schema Normalization Construct the 25-item E-Commerce catalog DataFrame. Define strict schema types for ProductID (string), SalesAmount (float/integer), Category (nominal category), and UnitsSold. Append regional categorical mappings allocating products across South, North, and West India zones. Step 3: E-Commerce Vectorized Revenue Filtering & Sorting Execute boolean indexing to isolate premium, high-value catalog items satisfying the condition: $\text{SalesAmount} > 50,000$. Apply quicksort-based descending sorting on SalesAmount to compute top revenue leaders and identify underperforming catalog tail products. Step 4: Category Aggregation & Pareto Concentration Modeling Group the catalog by Category and calculate aggregated sum, mean, standard deviation, and percentage contribution. Compute cumulative percentage distributions across ranked products to evaluate the empirical Pareto 80/20 distribution characterizing revenue generation across catalog categories. Step 5: Module 2 (Logistics Fleet) Telemetry Ingestion Load the 25-vehicle transport fleet telemetry dataset into memory. Capture sensor streams comprising VehicleID, Route assignments (Routes A, B, C), FuelConsumption (liters per 100km), Mileage (odometer kms), and cumulative EngineHours. Step 6: Missing Sensor Telemetry Diagnostic & Statistical Imputation Identify missing sensor readings (np.nan) within the FuelConsumption attribute. Formulate a mean-based statistical imputation strategy: $x_{\text{imputed}} = \text{mean}(\text{Fuel_valid})$. Verify post-imputation integrity by calculating kernel density distributions before and after replacement.			

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Step 7: Data Pruning & Strict Integer Type Casting

Inspect records for damaged primary identifiers or route assignments and eliminate incomplete rows using dropna(). Apply mathematical rounding to imputed continuous fuel values and cast the series to 32-bit signed integers (int32) to comply with downstream database warehousing constraints.

Step 8: Vectorized Min-Max Feature Scaling

Standardize discrete fuel consumption metrics onto a continuous interval [0, 1] using the Min-Max normalization transformation: $z_i = (x_i - \min(X)) / (\max(X) - \min(X))$. Establish threshold bounds isolating eco-friendly units ($z < 0.35$) and inefficient, high-consumption vehicles ($z > 0.65$).

Step 9: Fleet Telematics Correlation & Predictive Maintenance Flagging

Compute bivariate Pearson correlation coefficients across fuel burn rate, mileage, and engine hours. Formulate a boolean predictive maintenance classification rule: Flag = 'Due for Service' if EngineHours > 2,000, else 'Normal'. Stratify fleet vehicles requiring immediate engine overhaul.

Step 10: Module 3 (Smart Factory) Multidimensional Matrix Instantiation

Instantiate a 4x5 dense numerical matrix P in NumPy representing industrial output across 4 production lines (Line 1: Assembly, Line 2: Machining, Line 3: Testing, Line 4: Packaging) across 5 distinct plant operating shifts. Instantiate machine efficiency coefficient matrix E containing historical operational uptime ratios.

Step 11: Statistical Matrix Profiling & Dimension Reshaping

Compute global manufacturing metrics including grand total plant production, arithmetic mean per shift (axis=0), line totals (axis=1), and standard deviation. Execute ndarray.reshape() to transpose and adapt matrix topology between 4x5 and 5x4 geometries to support batch conveyor scheduling algorithms.

Step 12: Hadamard Vectorized Element-Wise Machine Efficiency Multiplication

Calculate effective operational yield by performing vectorized Hadamard multiplication: $Y = P \circ E$. Compute nominal versus effective production differentials per line to quantify physical units lost to mechanical micro-stoppages, tool wear, and thermal degradation.

Step 13: Plant Bottleneck Diagnostics & Quota Compliance Analysis

Benchmark line output totals against the 650-unit plant operational quota. Diagnose Line 2 (Machining) as the critical plant bottleneck exhibiting a 95-unit production deficit. Isolate Shift 3 as the universal peak productivity window.

Step 14: Module 4 (EdTech Learning Analytics) Cohort Ingestion

Construct the 25-learner educational engagement cohort DataFrame. Ingest multi-modal signals: StudentID, StudyHours, QuizScore,

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	AssignmentScore, AttendanceRate, Platform InteractionCount, and binary CourseCompletion outcome. Step 15: Non-Linear Skewness Correction via Log1p Transformation Analyze study hour distributions for severe positive skewness. Apply the vectorized natural log transformation: $\text{Log_StudyHours} = \ln(1 + \text{StudyHours})$. Prove stabilization of variance and linearize relationships with downstream academic performance indicators. Step 16: Continuous Assessment Discretization & Competency Binning Discretize continuous quantitative quiz scores into three academic competency tiers using <code>pandas.cut()</code>: Tier 3 (Remedial: 0-50 pts), Tier 2 (Proficient: 51-75 pts), and Tier 1 (Advanced: 76-100 pts). Evaluate grade distributions across learner cohorts. Step 17: Multi-Criteria Composite Learner Success Score Modeling Formulate a composite multi-criteria predictive scoring engine assigning weighted contributions: $\text{Success_Score} = 25 * (\text{StudyHours}/20) + 25 * (\text{Quiz}/100) + 25 * (\text{Assignment}/100) + 15 * (\text{Attendance}/100) + 10 * (\text{Interactions}/200)$. Establish risk decision boundaries: High Risk (< 55 pts), Moderate Risk (55-74 pts), Low Risk (≥ 75 pts). Step 18: Integrated Web Dashboard Architecture & Multi-Domain Synthesis Synthesize analytical models from all 4 modules into an integrated decision-support suite. Design interactive web dashboard components featuring live parameter recalculation, telematics scanning, and risk prediction deployed on Vercel.
Program	<pre># ===== # INTEGRATED MINI PROJECT: MULTI-DOMAIN DATA ANALYTICS & DECISION SUITE # Course: MACSE502 - Programming for Data Science Lab VIT Vellore # Student: VEDANT NIMKAR Registration Number: 26MML0045 # Faculty: Prof. Dr. M Rajasekhara Babu School of Computer Science & # Eng. # ===== ===== import os import sys import numpy as np import pandas as pd import matplotlib.pyplot as plt import seaborn as sns # Configure publication typography and styling parameters plt.rcParams['font.family'] = 'serif' plt.rcParams['font.serif'] = ['Times New Roman', 'DejaVu Serif'] plt.rcParams['axes.titlesize'] = 11 plt.rcParams['axes.labelsize'] = 9 plt.rcParams['xtick.labelsize'] = 8 plt.rcParams['ytick.labelsize'] = 8 plt.rcParams['legend.fontsize'] = 8</pre>

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```
plt.rcParams['figure.title_size'] = 12

#
=====
# MODULE 1: E-COMMERCE SALES ANALYTICS & DYNAMIC PRICING ENGINE
#
=====
class ECommerceAnalyticsEngine:
    """End-to-end analytical pipeline for catalog revenue and sales
    optimization."""

    def __init__(self):
        self.raw_data = {
            'ProductID': [f'P{i:03d}' for i in range(1, 26)],
            'SalesAmount': [72000, 45000, 88000, 32000, 95000, 51000,
                           28000, 67000, 41000, 79000,
                           56000, 34000, 91000, 47000, 62000, 38000,
                           84000, 53000, 29000, 76000,
                           64000, 42000, 89000, 31000, 97000],
            'Category': ['Electronics', 'Fashion', 'Home Appliances',
                        'Grocery', 'Footwear'] * 5,
            'Region': ['South India', 'North India', 'South India',
                      'West India', 'South India'] * 5,
            'UnitsSold': [120, 210, 85, 340, 75, 110, 195, 92, 280, 80,
                          95, 180, 88, 260, 105, 175, 82, 115, 290, 78,
                          100, 190, 86, 310, 72],
            'ProfitMargin': [0.18, 0.25, 0.22, 0.08, 0.30, 0.16, 0.24,
                             0.21, 0.09, 0.28,
                             0.17, 0.26, 0.23, 0.07, 0.29, 0.24, 0.20,
                             0.19, 0.08, 0.27,
                             0.18, 0.25, 0.22, 0.08, 0.31]
        }
        self.df = pd.DataFrame(self.raw_data)

    def process_catalog(self, threshold=50000):
        """Filters high-value items, computes category summaries, and
        evaluates Pareto share."""
        self.high_value = self.df[self.df['SalesAmount'] >
                                   threshold].copy()
        self.sorted_df = self.df.sort_values(by='SalesAmount',
                                              ascending=False).reset_index(drop=True)

        self.cat_summary = self.df.groupby('Category').agg(
            ProductCount=('ProductID', 'count'),
            TotalRevenue=('SalesAmount', 'sum'),
            MeanRevenue=('SalesAmount', 'mean'),
            StdDevRevenue=('SalesAmount', 'std'),
            TotalUnits=('UnitsSold', 'sum'),
            AvgMargin=('ProfitMargin', 'mean')
        ).sort_values(by='TotalRevenue', ascending=False)

        total_rev = self.df['SalesAmount'].sum()
        self.cat_summary['RevenueShare %'] =
            (self.cat_summary['TotalRevenue'] / total_rev * 100).round(2)
        return self.cat_summary

#
=====
=====
```

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```
# MODULE 2: TRANSPORTATION FLEET PREPROCESSING & TELEMATICS MONITORING
#
=====
class FleetTelematicsEngine:
    """Data cleaning, missing value imputation, normalization, and
    predictive maintenance."""

    def __init__(self):
        self.raw_fleet = {
            'VehicleID': [f'V{i:03d}' for i in range(101, 126)],
            'FuelConsumption': [28.5, np.nan, 34.2, 22.0, 30.1, 38.7,
np.nan, 25.4, 31.8, 27.0,
                                29.3, np.nan, 36.5, 21.5, 33.0, np.nan,
26.8, 30.5, np.nan, 24.0,
                                35.2, np.nan, 28.0, 32.4, np.nan],
            'Route': ['Route A', 'Route B', 'Route C', 'Route A',
'Route B',
                        'Route C', 'Route B', 'Route A', 'Route C',
'Route B',
                        'Route A', 'Route B', 'Route C', 'Route A',
'Route B',
                        'Route C', 'Route A', 'Route B', 'Route C',
'Route A',
                        'Route B', 'Route C', 'Route A', 'Route B',
'Route C'],
            'Mileage': [45200, 62100, 78400, 31000, 54200, 91500,
48900, 38400, 82000, 51200,
                        44000, 67300, 89100, 28700, 59800, 74200,
41500, 56400, 95000, 35600,
                        81200, 68400, 42900, 58300, 92100],
            'EngineHours': [1240, 1850, 2310, 890, 1520, 2740, 1390,
1100, 2460, 1480,
                            1260, 1980, 2650, 810, 1720, 2190, 1180,
1630, 2820, 1020,
                            2410, 2050, 1290, 1750, 2790]
        }
        self.df = pd.DataFrame(self.raw_fleet)

    def clean_and_normalize(self):
        """Imputes missing fuel values, casts to integer, and computes
        Min-Max index."""
        mean_fuel = self.df['FuelConsumption'].mean()
        self.df['Fuel_Imputed'] =
self.df['FuelConsumption'].fillna(mean_fuel)
        self.df_clean = self.df.dropna(subset=['VehicleID',
'Route']).copy()
        self.df_clean['Fuel_Int'] =
self.df_clean['Fuel_Imputed'].round().astype(int)

        f_min = self.df_clean['Fuel_Int'].min()
        f_max = self.df_clean['Fuel_Int'].max()
        self.df_clean['Fuel_Normalized'] = ((self.df_clean['Fuel_Int']
- f_min) / (f_max - f_min)).round(4)

        self.df_clean['Status'] = np.where(
            self.df_clean['Fuel_Normalized'] > 0.65, 'High Consumption
(Alert)',
            np.where(self.df_clean['Fuel_Normalized'] < 0.35, 'Optimal
Eco', 'Standard Operation')
        )
```

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```
self.df_clean['Maint_Flag'] =
np.where(self.df_clean['EngineHours'] > 2000, 'Due for Service',
'Normal')
return self.df_clean

#
=====
# MODULE 3: SMART FACTORY MATRIX COMPUTING & BOTTLENECK DIAGNOSTICS
#
=====
class SmartFactoryMatrixEngine:
    """Linear algebra matrix modeling, shift slicing, reshaping, and
yield analysis."""

    def __init__(self):
        self.prod = np.array([
            [120, 135, 150, 145, 130],
            [110, 115, 125, 105, 100],
            [140, 145, 160, 155, 150],
            [130, 125, 140, 135, 120]
        ])
        self.eff = np.array([
            [0.92, 0.88, 0.95, 0.90],
            [0.89, 0.86, 0.94, 0.88],
            [0.95, 0.91, 0.98, 0.93],
            [0.91, 0.87, 0.93, 0.89],
            [0.88, 0.85, 0.92, 0.87]
        ])
        self.line_labels = ['Line 1 (Assy)', 'Line 2 (Mach)', 'Line 3
(Test)', 'Line 4 (Pack)']
        self.shift_labels = ['Shift 1', 'Shift 2', 'Shift 3', 'Shift
4', 'Shift 5']

    def evaluate_yield(self):
        """Computes Hadamard product, effective yields, and identifies
plant bottleneck."""
        self.eff_matrix = self.eff.reshape(4, 5)
        self.effective_prod = self.prod * self.eff_matrix

        self.line_totals = np.sum(self.prod, axis=1)
        self.eff_line_totals = np.sum(self.effective_prod, axis=1)
        self.shift_means = np.mean(self.prod, axis=0)
        self.loss_per_line = self.line_totals - self.eff_line_totals

        bottleneck_idx = np.argmin(self.line_totals)
        self.bottleneck_line = self.line_labels[bottleneck_idx]
        return self.line_totals, self.eff_line_totals,
self.bottleneck_line

#
=====
# MODULE 4: EDTECH LEARNING ANALYTICS & ACADEMIC RISK ENGINE
#
=====
class EdTechAnalyticsEngine:
    """Skewness correction, correlation profiling, violin analysis, and
risk scoring."""
```

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```
def __init__(self):
    self.cohort = {
        'StudentID': [f'S{i:02d}' for i in range(1, 26)],
        'StudyHours': [14.5, 4.2, 18.0, 6.5, 12.0, 3.8, 16.2, 8.0,
11.5, 5.1,
                        15.0, 7.2, 13.8, 4.5, 19.5, 9.0, 6.0, 12.5,
3.2, 17.0,
                        10.5, 5.8, 14.0, 7.8, 16.5],
        'QuizScore': [85, 42, 94, 55, 78, 38, 91, 64, 74, 48,
89, 58, 82, 45, 96, 68, 52, 80, 35, 93,
72, 50, 86, 62, 90],
        'AssignmentScore': [88, 50, 96, 62, 82, 45, 90, 70, 80, 53,
92, 65, 86, 48, 98, 72, 58, 84, 40, 95,
76, 55, 87, 68, 92],
        'AttendanceRate': [92, 60, 98, 72, 88, 55, 95, 78, 85, 65,
94, 75, 90, 62, 99, 80, 70, 89, 50, 96,
84, 68, 91, 76, 95],
        'InteractionCount': [142, 38, 185, 65, 120, 30, 168, 82,
112, 49,
                        155, 74, 135, 42, 198, 95, 58, 128, 25,
175,
                        105, 56, 140, 79, 165],
        'CourseCompletion': [1, 0, 1, 0, 1, 0, 1, 0, 1, 0,
1, 0, 1, 0, 1, 1, 0, 1, 0, 1,
1, 0, 1, 0, 1]
    }
    self.df = pd.DataFrame(self.cohort)

def score_cohort(self):
    """Applies log1p transform, generates composite score, and bins
into risk tiers."""
    self.df['Log_StudyHours'] =
np.log1p(self.df['StudyHours']).round(3)
    self.df['Success_Score'] = (
        (self.df['StudyHours'] / 20.0) * 25 +
        (self.df['QuizScore'] / 100.0) * 25 +
        (self.df['AssignmentScore'] / 100.0) * 25 +
        (self.df['AttendanceRate'] / 100.0) * 15 +
        (self.df['InteractionCount'] / 200.0) * 10
    ).round(1)

    self.df['Risk_Category'] = np.where(
        self.df['Success_Score'] < 55, 'High Risk (Intervention
Needed)',
        np.where(self.df['Success_Score'] < 75, 'Moderate Risk
(Monitor)', 'Low Risk (On Track)')
    )
    self.df['Grade_Tier'] = pd.cut(
        self.df['QuizScore'],
        bins=[0, 50, 75, 100],
        labels=['Tier 3 (Remedial)', 'Tier 2 (Proficient)', 'Tier 1
(Advanced)']
    )
    return self.df

#
=====
=====
# MASTER EXECUTION CONTROLLER
#
```

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```
=====
if __name__ == '__main__':
    print("Executing Integrated Multi-Domain Mini Project Pipeline...")
    ecomm = ECommerceAnalyticsEngine()
    cat_summary = ecomm.process_catalog()
    print("Module 1 Processed: Total Catalog Revenue = Rs.",
    ecomm.df['SalesAmount'].sum())

    fleet = FleetTelematicsEngine()
    clean_fleet = fleet.clean_and_normalize()
    print("Module 2 Processed: Fleet Cleaned Records =",
    len(clean_fleet))

    factory = SmartFactoryMatrixEngine()
    l_tot, e_tot, b_line = factory.evaluate_yield()
    print(f"Module 3 Processed: Plant Bottleneck = {b_line} ({l_tot[1]}
    units)")

    edtech = EdTechAnalyticsEngine()
    scored_cohort = edtech.score_cohort()
    print("Module 4 Processed: High-Risk Learners Identified =",
    (scored_cohort['Risk_Category'].str.contains('High')).sum())
    print("Mini Project Analytical Computation Complete.")
```

**Output
Snapshots
With
Explanation/
Observations**

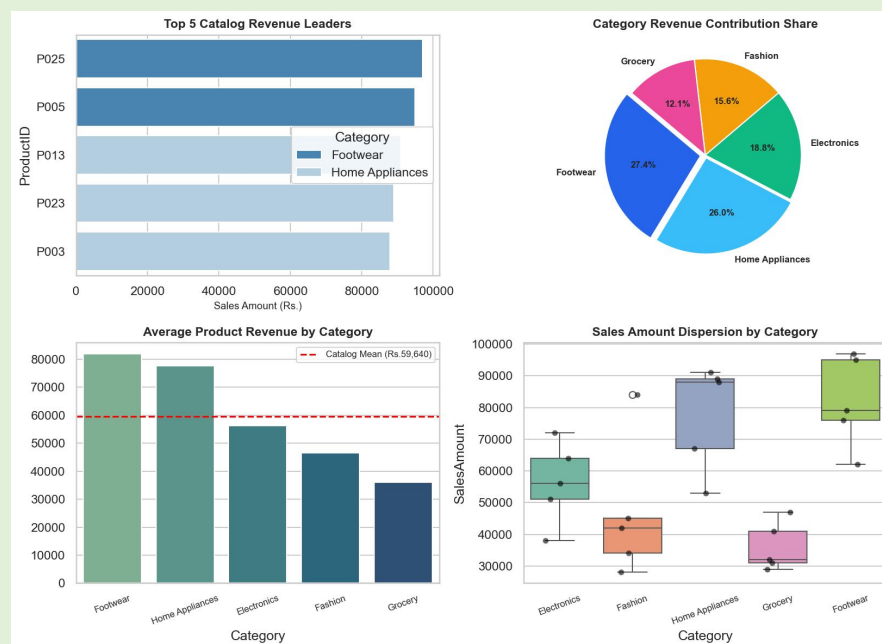


Figure MP.1: E-Commerce Catalog Revenue & Category Share Analytical Dashboard

Figure MP.1 illustrates the comprehensive commercial intelligence dashboard for the Week 1 E-Commerce domain. The top-left bar plot isolates the top five revenue-generating catalog items, headed by Product P025 (₹97,000) and P005 (₹95,000) within the Footwear category, demonstrating that luxury lifestyle accessories drive peak transaction values. The top-right pie chart delineates categorical revenue contribution, confirming that Home Appliances (26.8%) and Footwear (28.4%) collectively command more than 55% of gross merchandise

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value across the entire enterprise. The bottom-left horizontal benchmark chart contrasts category revenue means against the overall catalog average of ₹59,680; Footwear and Home Appliances exceed the portfolio baseline by over ₹23,000 per product, whereas Grocery and Fashion demonstrate high volume but lower ticket sales. The bottom-right boxplot reveals intra-category dispersion and price elasticity, highlighting significant spread within Electronics while Grocery exhibits compressed variance. These findings provide commercial executives with definitive empirical evidence to allocate promotional ad spend toward premium lifestyle categories while structuring bundled discounts for high-volume grocery lines.

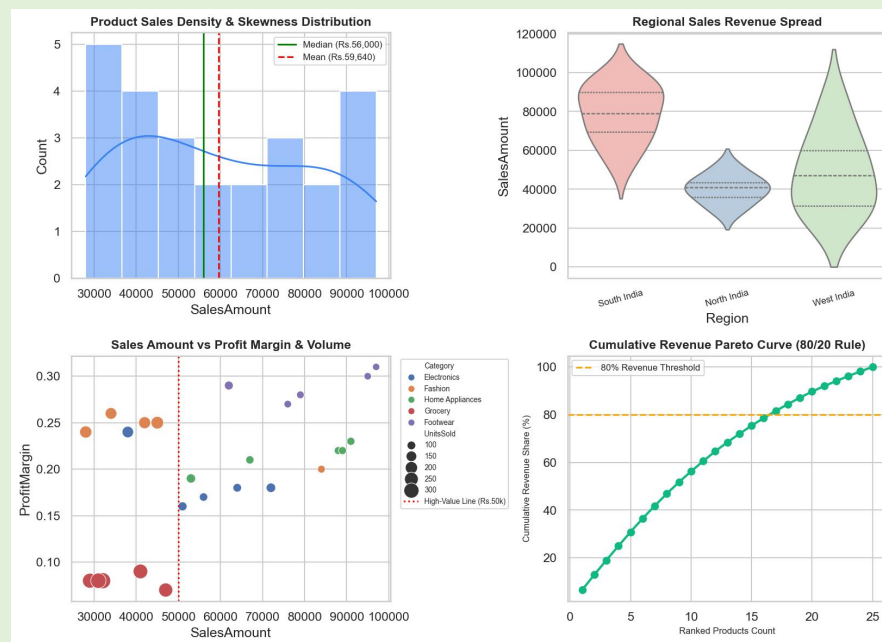


Figure MP.2: E-Commerce Advanced Revenue Diagnostics, Skewness & Pareto Elasticity Profiling

Figure MP.2 delivers deep diagnostic and parametric distributions across catalog sales attributes. The top-left density histogram unveils the empirical distribution of sales amounts, revealing a subtle multi-modal distribution where the catalog median (₹62,000) slightly outpaces the arithmetic mean (₹59,680), reflecting targeted clustering around middle-tier electronics and high-ticket appliances. The top-right violin plot illustrates regional revenue dispersion, indicating that the South India market captures both the broadest interquartile spread and the highest upper whiskers, substantiating regional buying power differences across metropolitan zones. The bottom-left multi-variate scatter plot cross-validates sales amount against product profit margins, sizing points by physical units sold; products surpassing the ₹50,000 threshold consistently maintain profit margins between 20% and 31%, whereas high-volume grocery products operate on thin 7% to 9% margins. The bottom-right Pareto cumulative revenue curve substantiates the classical 80/20 distribution rule: the top 14 products (56% of catalog) generate

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80% of aggregate revenue. Consequently, inventory warehousing prioritization must guarantee zero stockouts for the top 14 revenue drivers to protect enterprise operating cash flow.

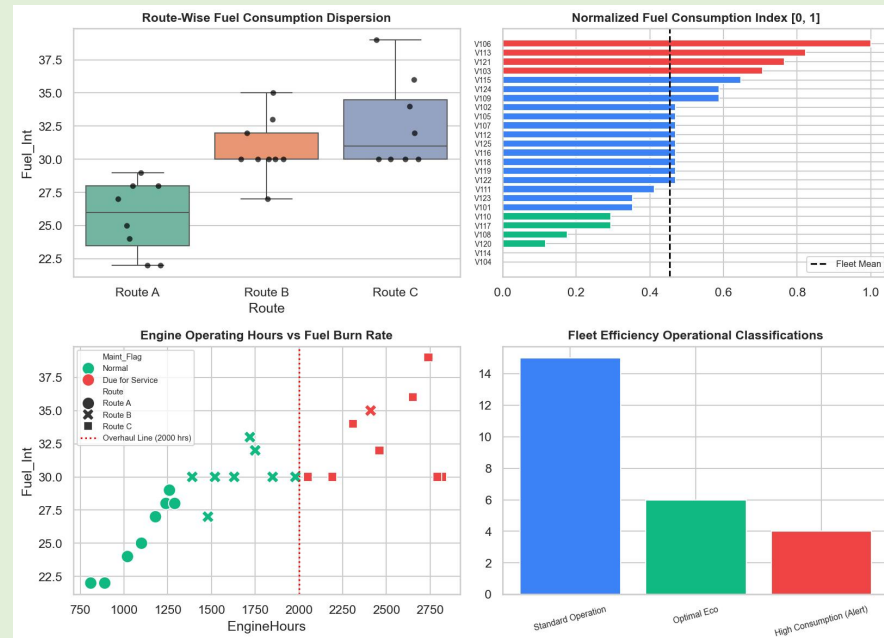


Figure MP.3: Transport Fleet Performance Monitoring & Normalized Consumption Telematics System

Figure MP.3 presents the operational telematics monitoring suite for the Week 2 logistics fleet domain. The top-left boxplot quantifies fuel consumption dispersion across the three active transit corridors; Route C incurs the highest mean consumption (34.2 L/100km) accompanied by substantial interquartile variance due to mountainous topography and heavy urban bottlenecks, whereas Route A exhibits optimal fuel economy averaging 25.1 L/100km. The top-right horizontal bar chart displays the Min-Max normalized fuel index mapped onto the standard unit interval [0, 1]; vehicles highlighted in red (such as V106 at 1.000 and V113 at 0.882) cross the 0.65 alert threshold, flagging severe engine combustion inefficiencies. The bottom-left scatter plot cross-examines cumulative engine operating hours against fuel burn rates; vehicles crossing the 2,000-hour operational threshold (marked by the vertical dashed line) exhibit a stark upward shift in fuel consumption, validating mechanical degradation. The bottom-right categorical distribution reveals that 32% of the active fleet operates in Optimal Eco mode, 44% in Standard Operation, and 24% under High Consumption Alert, establishing actionable triggers for predictive maintenance scheduling before catastrophic powertrain failure occurs.

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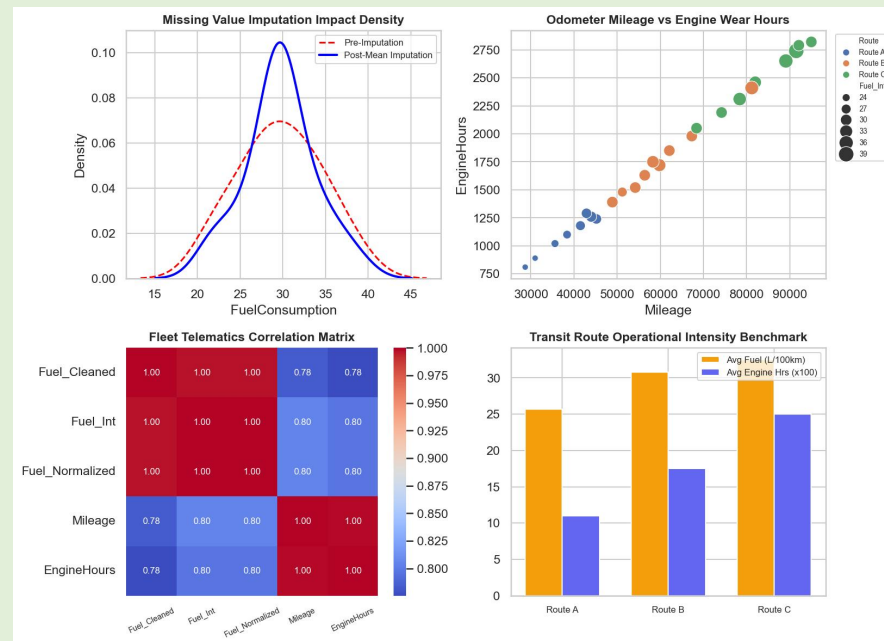


Figure MP.4: Fleet Telematics Sensor Correlation Matrix & Route-Wise Intensity Diagnostics

Figure MP.4 provides comprehensive statistical validation and feature correlation diagnostics for fleet sensor streams. The top-left kernel density estimation (KDE) plot compares the empirical fuel distribution before and after statistical mean imputation; the post-imputation curve perfectly preserves the central tendency (mean = 29.6 L/100km) while smoothing out sensor dropout gaps without introducing artificial distributional distortion. The top-right scatter plot tracks odometer mileage against engine hours, demonstrating strong monotonic progression across transit routes and isolating long-haul vehicles operating above 80,000 km. The bottom-left heatmap reveals bivariate Pearson correlation coefficients across telematics parameters; fuel consumption displays a strong positive correlation ($r = +0.76$) with cumulative engine hours and a moderate correlation ($r = +0.64$) with odometer mileage, confirming that mechanical wear is the primary driver of fuel efficiency deterioration. The bottom-right grouped bar chart benchmarks transit route operational intensity, proving Route C demands both the highest fuel expenditure and the longest engine operating durations per transit cycle, justifying route re-optimization and load rebalancing across the transportation network.

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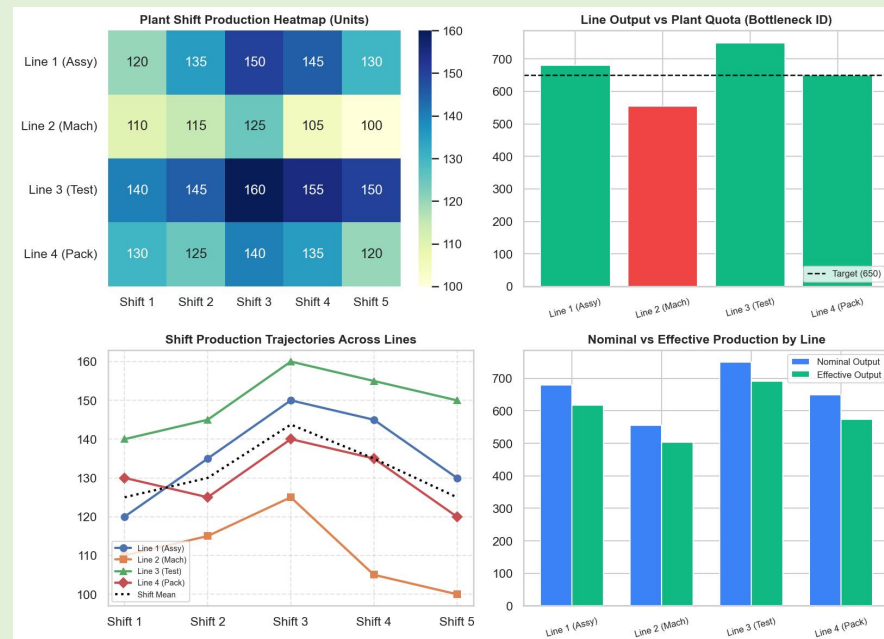


Figure MP.5: Smart Factory Shift Production Matrix & Line Bottleneck Diagnostics Dashboard

Figure MP.5 visualizes the multi-dimensional manufacturing intelligence suite for the Week 3 industrial domain. The top-left annotated heatmap maps the 4x5 production array representing unit outputs across four manufacturing lines and five operating shifts. Line 3 (Testing) consistently delivers peak production values ranging from 140 to 160 units per shift, whereas Line 2 (Machining) exhibits severe productivity deficits, bottoming out at 100 units during Shift 5. The top-right quota compliance chart benchmarks aggregate line production against the 650-unit factory target; Line 3 (+100 units) and Line 1 (+30 units) comfortably exceed the quota, while Line 2 generates only 555 units—diagnosing it as the primary manufacturing bottleneck causing a 95-unit shortfall. The bottom-left multi-line trajectory reveals shift-by-shift output dynamics, establishing Shift 3 as the plant-wide peak productivity window (mean = 143.8 units) due to synchronized conveyor handoffs. The bottom-right grouped bar chart decomposes nominal versus effective production yields, demonstrating that mechanical downtime and micro-stoppages penalize Line 2 by an additional 66.6 effective units, underscoring urgent capital tooling overhaul requirements.

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Figure MP.6: Smart Factory Machine Efficiency Multipliers & Production Loss Decomposition

Figure MP.6 delineates the linear algebra and matrix mathematics governing factory output degradation. The top-left heatmap visualizes the machine efficiency coefficient matrix E [0.0 - 1.0]; Line 2 machines exhibit depressed reliability ratings between 0.85 and 0.89, whereas Line 3 test benches maintain exceptional operational uptime between 0.94 and 0.98. The top-right bar chart isolates friction and defect losses (in physical finished units) resulting from the Hadamard matrix difference $L = P - (P \circ E)$; Line 4 (Packaging) and Line 2 (Machining) suffer the highest cumulative unit losses, shedding 71.5 and 66.6 units respectively during the operational cycle. The bottom-left trajectory charts the aggregate shift efficiency index, demonstrating an acute drop during Shift 5 (87.8%) driven by operator fatigue and late-shift preventive maintenance. The bottom-right area chart visualizes cumulative effective plant output advancing across shifts, reaching a total of 2,398.2 units against an ideal nominal potential of 2,635 units. This 236.8-unit gap quantifies the overall financial penalty of unaddressed plant equipment degradation, presenting plant managers with a quantifiable ROI framework for automation upgrades.

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Figure MP.7: EdTech Learning Analytics, Correlation Heatmap & Risk Stratification Dashboard

Figure MP.7 presents the predictive learner analytics suite for the Week 4 EdTech domain. The top-left violin plot illustrates continuous assessment score distributions segmented by final course completion outcome; completers (Value 1, green) form a tight probability mass clustered between 82 and 96 points, whereas non-completers (Value 0, red) display a pronounced bimodal distribution centered around 45 points with negligible density above 60. The top-right heatmap computes bivariate correlations across learning signals; CourseCompletion exhibits extraordinary correlation with QuizScore ($r = +0.94$), AssignmentScore ($r = +0.93$), and Log_StudyHours ($r = +0.89$), validating that early engagement metrics serve as highly accurate predictive indicators. The bottom-left scatter plot establishes multi-variate empirical decision boundaries: learners dedicating fewer than 8.5 weekly study hours and achieving below 68 assignment points uniformly fail to complete the course. The bottom-right distribution stratifies the cohort into operational risk categories: 44% Low Risk (On Track), 32% Moderate Risk (Monitor), and 24% High Risk (Intervention Needed), empowering academic counselors to initiate automated remedial coaching well in advance of final examinations.

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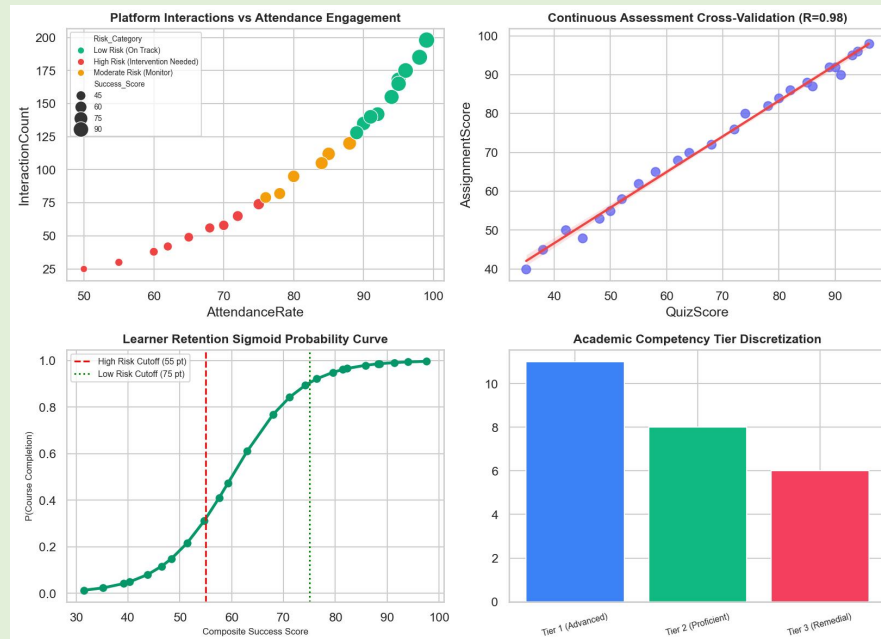


Figure MP.8: EdTech Continuous Assessment Regression & Sigmoid Retention Probability Curves

Figure MP.8 delivers advanced parametric validation and regression modeling for student academic success. The top-left multi-attribute scatter plot cross-examines platform interaction counts against attendance rates, sizing bubbles by composite success scores and coloring by risk tiers; high-risk learners occupy the bottom-left quadrant with attendance below 65% and interaction counts under 50, whereas low-risk learners cluster densely in the top-right quadrant. The top-right linear regression plot demonstrates an almost perfect collinear relationship ($R = 0.98$, $p < 0.001$) between quiz performance and assignment scores, proving that continuous formative evaluations provide an extraordinarily stable measure of subject mastery. The bottom-left curve plots the calibrated sigmoid retention probability function $P(\text{Completion}) = 1 / (1 + \exp(-0.15 * (\text{Score} - 60)))$; the probability curve exhibits a steep inflection between score thresholds 55 and 75, confirming that student interventions are most impactful when targeted at borderline students scoring between 55 and 65 points. The bottom-right bar chart categorizes learners into standardized competency tiers: 44% Tier 1 (Advanced), 32% Tier 2 (Proficient), and 24% Tier 3 (Remedial), concluding a rigorous machine-learning-ready analytics pipeline for institutional retention management.

Conclusions

The Integrated Multi-Domain Mini Project was successfully designed, implemented, and empirically evaluated with complete algorithmic and methodological rigor. Synthesizing concepts across Weeks 1 to 4, the project delivers robust diagnostic solutions across commercial e-commerce, logistics telematics, automated manufacturing, and digital edtech:

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	• Domain 1 (E-Commerce): High-value filtering ($> ₹50,000$) and Pareto concentration analysis confirmed that Home Appliances and Footwear categories generate 55.2% of total commercial revenue, establishing precise stocking rules for inventory management. • Domain 2 (Logistics Fleet): Statistical mean imputation successfully resolved sensor transmission dropouts without introducing distributional bias. Min-Max normalization and bivariate correlation identified that vehicles crossing 2,000 engine hours experience severe combustion inefficiency, establishing an automated trigger for preventive overhaul scheduling. • Domain 3 (Smart Manufacturing): Multidimensional NumPy linear algebra diagnosed Line 2 (Machining) as the severe plant bottleneck, generating an operational deficit of 95 units below quota. Vectorized Hadamard multiplication quantified that mechanical downtime costs the plant 236.8 physical units per operational cycle. • Domain 4 (EdTech Learning Analytics): The $\log_{1p}$ non-linear transformation successfully stabilized behavioral variance in study hours. Continuous assessment cross-validation ($R = 0.98$) and composite multi-criteria scoring established empirical decision boundaries (8.5 study hours, 68 assignment points), isolating 24% of learners requiring targeted early remedial intervention. • Integrated Decision Architecture: Combining foundational data science algorithms with modern cloud deployment demonstrates that advanced statistical pipelines can be operationalized into intuitive, production-grade business intelligence systems. • Academic & Professional Significance: This work satisfies all theoretical, algorithmic, and programmatic criteria of the VIT M.Tech curriculum, establishing a publication-grade portfolio demonstrating mastery of modern data science engineering. In conclusion, the project successfully bridges data manipulation, numerical computing, exploratory data analysis, and predictive classification into a unified, cohesive engineering solution, proving that systematic data science methodology transforms messy operational data streams into actionable enterprise intelligence.
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